Productizing an AI Data Product: A Report Pipeline

#Productization #Pipeline #Cross-functional collaboration #AI #0to1

Context

At Pebblous, I worked as a UX & Product lead for Data Clinic, a product that evaluates the quality of large, unstructured data for AI algorithms. Leadership wanted to see the evaluation outcome in the form of reports within 2 months — at least one full version of an evaluation report to showcase to investors — but there was no dedicated system to deliver the report at the time.

Challenges

- The process of producing an evaluation report for the sample datasets had not started even after a couple of months, and the delay was impacting other teams' productivity, particularly the work on the end-user part.
- The ML engine was ready to calculate the datasets' properties and numerical values, but the DS team had no clear way — and not enough resources — to frame those values from the perspective of evaluating data quality and into the form of a report.
- Lack of infrastructure: no pipeline was established to produce the report, and there was no clear structure for it.
- Tight timeline: given the DS and ML teams' schedules, the project would add a heavy burden on their work.
- Leadership had strongly urged the DS team, but the progress had not changed much.

Actions

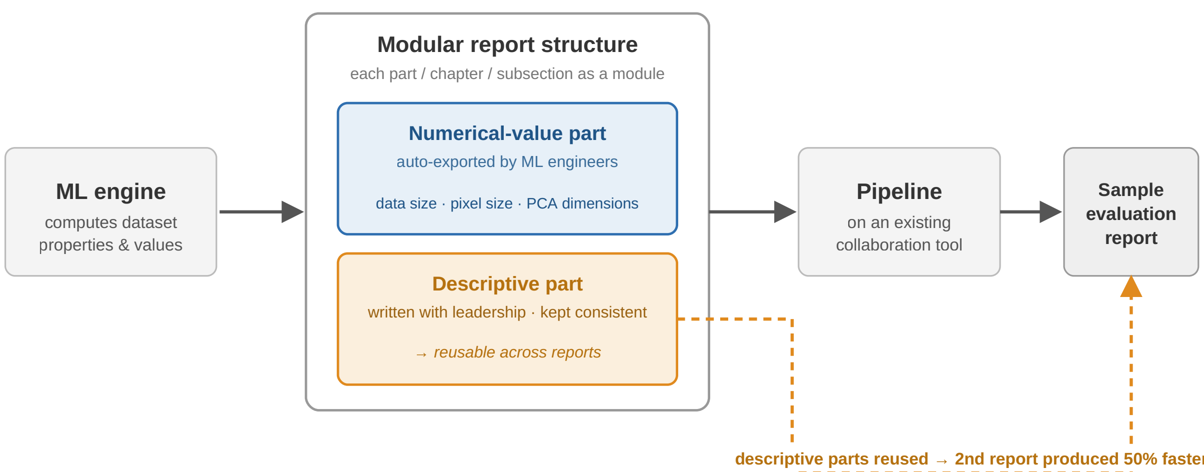
I proactively approached the DS team leader for a 1:1. Frustrated, she shared the issues blocking progress — although the ML engine could calculate the datasets' properties, the team didn't know how to frame the values in the form of a report, and they lacked the resources to figure it out. Through the conversation, I clarified that the real issue was a lack of guideline, not the team's productivity itself. As a product lead, I felt a strong sense of ownership and decided to take initiative to guide the DS team so they could organize the values within the frame of an end-user report.

Aligning the team. I launched a kick-off meeting to align everyone on the goal. I confirmed the target we wanted to achieve, clarified how much workload the DS and ML teams had now and in the near future, and learned the pain points preventing them from prioritizing the project. I set a clear minimal baseline — a system capable of delivering at least one sample report within 2 months — and decided to automate the system for future scale-up.

Choosing a resource-light approach. Considering the scarce resources and the tight timeline, I adopted an existing commercial collaboration tool instead of asking the engineering team to build an internal one, in order to save their resources. I finished the tool setup to make it easier for team members to access.

Designing and modularizing the report. I designed and organized the overall structure of the report, structuring the table of the report together with leadership. I modularized each part, chapter, and subsection, and within each module I separated the numerical-value parts from the written descriptive parts. I asked the ML engineers to directly export the numerical values onto the report — for example, data size, pixel size, and PCA dimensions — and I wrote the descriptive parts together with leadership, keeping them consistent to minimize updates.

Below is an overview of the report production pipeline.



Clarifying scope and keeping momentum. I clearly communicated the plan to both teams to assure them of what we could achieve, which reduced their psychological burden in taking on the new project. I clarified the scope and tasks — the teams would run the existing pipeline and push the outcome through the pipeline I was going to build — and I ran weekly meetings to help team members keep up, sharing weekly progress and clarifying what should be done in the next week.

Results

- My action re-motivated the DS team, and we finally developed a pipeline to produce the reports.
- The first sample report was produced after about a month — earlier than expected.
- During the second month, we iterated on the pipeline and the structure to produce the sample report even faster. Because the written descriptive parts were done and reused, the overall speed was greatly improved — once the descriptive parts were complete and the first report was produced, the time spent on the second report was reduced by 50%.
- We produced 3 sample reports by the middle of the second month.

Impact

- Leadership was able to showcase 3 sample reports — three different types and qualities of datasets — to investors and potential customers in a timely manner, and received positive feedback.
- The increased versatility of the reports positively impacted the next round of investment and helped secure an important business partnership.